

17 A aim is the NOT distinguish the pixels

hence use the smallest wavelength to determine the shortest pixel distance that can be distinguished.

$$\sin \theta \approx \frac{\lambda}{b} \approx \frac{s}{r}$$

for maximum distance between pixels $s_{\max}$, we consider shortest wavelength (blue):

$$\frac{d}{0.1} < \frac{\lambda}{b}$$

$$\frac{d}{0.1} < \frac{470 \times 10^{-9}}{4.0 \times 10^{-3}}$$

$$d < 1.2 \times 10^{-5} \text{ m}$$

18 B connecting wire so all metallic surfaces at same potential